

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

INSTITUTE / COLLEGE OF ENGINEERING & TECHNOLOGY

(Affiliated to State Technical University / AICTE Approved)

MINI PROJECT SYNOPSIS

SKYCAST AI

Autonomous Hyperlocal Weather Intelligence & Multi-Source Disaster Crisis
Command System

A Project Synopsis submitted in partial fulfillment of the academic curriculum requirements for the award of the degree of

BACHELOR OF TECHNOLOGY (B.TECH) / DIPLOMA

in **COMPUTER SCIENCE & ENGINEERING / INFORMATION TECHNOLOGY**

SUBMITTED BY (STUDENT):

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Roll No / Enrollment: [Your University Roll Number]

Semester / Year: 7th Semester / Final Year

Branch: Computer Science & Engineering

UNDER THE GUIDANCE OF:

Project Guide: [Prof. / Dr. Guide Name]

Designation: Assistant Professor / HOD

Department: Dept. of Computer Science & Engg.

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ACADEMIC SESSION: 2026 – 2027

1. PROJECT ABSTRACT

Traditional weather applications offer generic regional forecasts without actionable disaster response capabilities. During severe meteorological emergencies (floods, cloudbursts, severe lightning strikes, extreme heatwaves, and toxic smog), civilian casualties and administrative paralysis occur due to communication tower failures, lack of localized vulnerability data, and uncoordinated rescue team dispatch.

SkyCast AI is an autonomous, defense-grade **Hyperlocal Weather Intelligence and Multi-Source Disaster Crisis Management System** engineered for District Disaster Management Authorities (DDMA), National/State Disaster Response Forces (NDRF/SDRF), Gram Panchayats, and civic populations. The platform seamlessly synthesizes **Satellite Precipitation Radar (INSAT/Sentinel-3)**, **IoT River Level Telemetry**, **Pre-Registered Village Risk Databases**, and **Autonomous UAV Drone Reconnaissance (Thermal FLIR)** into a centralized command interface.

Featuring a **District AI Digital Twin Simulator** for flood inundation mapping, a **Thermodynamic CAPE Lightning Radar** with safe shelter routing, and a zero-network **Passive Citizen Disaster Geo-Anchor**, SkyCast AI transitions district emergency governance from a reactive scramble to a proactive, predictive, and zero-casualty operational standard.

2. PROBLEM STATEMENT (समस्या विवरण)

District administrations face critical systemic bottlenecks during climate emergencies:

- **Lack of Hyperlocal Granularity:** Existing national weather APIs report macroscopic city averages, failing to pinpoint inundation risk across specific tehsils, wards, or riverbank villages.
- **Communication & Cellular Blackouts:** Natural disasters frequently sever cellular base stations and power grids, rendering cloud-dependent emergency apps entirely inoperable.
- **Unscientific Rescue Resource Allocation:** NDRF/SDRF first responders lack a real-time mathematical prioritization matrix to determine which stranded populations (infants, elderly, flooded rooftops) require immediate boat or ambulance dispatch.
- **High Rural Lightning Mortality:** In convective weather, rural agricultural workers lack immediate thermodynamic early warning and distance-calculated routing to verified lightning-safe concrete shelters.

3. PROJECT OBJECTIVES

The primary engineering deliverables of the SkyCast AI project are:

1. **Autonomous Hyperlocal Forecasting:** To deliver real-time weather analytics across temperature, humidity, atmospheric pressure, wind gusts, US-AQI, and UV radiation with zero API key dependencies.

2. **District AI Digital Twin Simulation:** To construct an interactive topographical 2D/3D mesh simulator predicting flood surface area, submerged bridges, and generating safe green evacuation corridors for 0–250 mm rainfall stress tests.
3. **Thermodynamic Lightning Radar:** To compute convective atmospheric instability via CAPE and Lifted Index (LI), triggering early warnings 20–35 minutes prior to strikes and mapping nearby Grade-A shielded concrete shelters.
4. **Multi-Source Disaster Detection & Drone Reconnaissance:** To integrate satellite radar, IoT river gauges, and autonomous UAV canvas simulations with Thermal FLIR infrared imaging for detecting human heat signatures on submerged rooftops.
5. **Passive Citizen Disaster Geo-Anchoring:** To automatically cache a citizen's coordinates upon a single visit, matching their sector against real-time flood thresholds to enable zero-click GPS beacon distress dispatch during network blackouts.
6. **Smart NDRF Rescue Triage:** To implement an automated incident queue assigning appropriate response units (Team Alpha - Boats, Bravo - Paramedics, Charlie - Debris Clearance) based on vulnerability.

4. PROPOSED SYSTEM ARCHITECTURE

SkyCast AI is structured as a high-performance, modular client-side Single Page Application (SPA) and Progressive Web App (PWA), ensuring microsecond latency and offline resilience.

- System Architectural Pipeline:
- Telemetry Ingestion: Open-Meteo REST Endpoints + WMO Code Matrix + Satellite Radar Buffers
- Analytical Core: CAPE Instability Engine + AI Priority Scoring Matrix + Digital Twin Physics
- Visualization Layer: Leaflet.js Geospatial Engine + HTML5 Canvas 2D Topographic & UAV HUDs
- Resilience & Comms: Service Worker Cache + Web Audio Siren + VHF Radio Chirp + WhatsApp Gateway

Core Functional Modules:

- Module 1: Executive DDMA Command Center:** Comprehensive situational awareness console providing live IST master clocks, 24x7 control indicators, tehsil-level threat matrices, and vulnerable population directives.
- Module 2: District Digital Twin Simulator:** Topographical canvas simulating rising flood inundation levels (0.018 m/mm), submerged infrastructure count, and computing dynamic bypass corridors.
- Module 3: Multi-Source Detection & Drone HUD:** Sensor fusion interface featuring live toggleable FLIR Thermal (0.0518 spectrum) and 4K RGB daylight drone cameras with active target reticles.
- Module 4: Passive Citizen Disaster Geo-Anchor:** LocalStorage persistent beacon automatically verifying citizen safety margins and generating formatted Google Maps GPS distress links.
- Module 5: Smart Rescue Operations (NDRF/SDRF):** Live battalion readiness tracking, capacity management, equipment chips, and a field check-in modal.
- Module 6: Zero-Network Alternative Comms Hub:** Integration with VHF Marine/Emergency CH-16 (156.800 MHz), satellite handheld dispatcher, and offline queued SOS synchronization.
- Module 7: Bilingual Executive Bulletin Generator:** 1-click generation and print-ready export of official DDMA disaster bulletins in English and Hindi.

5. LITERATURE SURVEY & COMPARATIVE ANALYSIS

FEATURE / CAPABILITY	STANDARD WEATHER APPS (ACCUWEATHER/WINDY)	GOVT PORTALS (IMD/NDMA)	SKYCAST AI (PROPOSED SYSTEM)
Hyperlocal Focus	City Level Only	District Averages	Tehsil, Ward & Village Level

FEATURE / CAPABILITY	STANDARD WEATHER APPS (ACCUWEATHER/WINDY)	GOVT PORTALS (IMD/NDMA)	SKYCAST AI (PROPOSED SYSTEM)
Digital Twin Inundation	✗ Not Available	✗ Static PDF Maps	✓ Real-Time 2D/3D Interactive
Drone Thermal FLIR HUD	✗ Not Available	✗ Separate Manual Ops	✓ Integrated Aerial HUD Simulation
Offline & Network Resilience	✗ Fails without 4G/5G	✗ Server-dependent	✓ 100% PWA Offline SOS Queue
Passive Geo-Anchor Caching	✗ Session Only	✗ Requires Aadhaar Login	✓ 1-Visit Auto Life-Safety Cache
NDRF Autonomous Triage	✗ None	Manual Phone Logs	✓ AI Multi-Factor Priority Score
Language Support	English Primarily	Bilingual (Static)	✓ Dynamic 1-Click English ⇌ Hindi

6. MATHEMATICAL MODELING & ALGORITHMIC FORMULATION

A. AI Rescue Priority Score (Ψ_{rescue}):

To prevent triage bias, first-responder deployment priority is determined via a multi-factor linear weighting equation:

$$\Psi_{\text{rescue}} = (F_{\text{rain}} \cdot 0.30) + (R_{\text{surge}} \cdot 0.25) + (P_{\text{vuln}} \cdot 0.20) + (H_{\text{hist}} \cdot 0.15) + (D_{\text{drone}} \cdot 0.10)$$

- Where:
- F_{rain} : Simulated rainfall inundation factor [0-100]
 - R_{surge} : River gauge surge above danger threshold (e.g. +3.4m)
 - P_{vuln} : Vulnerability ratio (Infants & Elderly / Total Population)
 - H_{hist} : Historical inundation frequency weight
 - D_{drone} : UAV FLIR thermal human heat signature confirmation score

B. Convective Available Potential Energy (CAPE) Lightning Index:

$$\text{CAPE} = \int_{z_{\text{LFC}}}^{z_{\text{EL}}} g \left(\frac{T_{\text{v, parcel}}}{T_{\text{v, env}}} - 1 \right) dz$$

- Threshold Directive:
- $\text{CAPE} < 1000 \text{ J/kg}$: Low / Stable (Green Pill)
 - $1000 \leq \text{CAPE} < 2000 \text{ J/kg}$: Moderate Thunderstorm Threat (Yellow Pill)
 - $\text{CAPE} \geq 2000 \text{ J/kg}$: Severe Lightning & Cloud-to-Ground Strike Red Alert

7. HARDWARE & SOFTWARE REQUIREMENTS

CATEGORY	SPECIFICATION / TECHNOLOGY STACK
Frontend Architecture	HTML5, Advanced Modern CSS3 (Glassmorphism, Flexbox, CSS Grid, Media Queries)
Programming Language	Pure Vanilla JavaScript (ECMAScript 2022+), Zero Heavy Framework Overhead
Graphics & Simulation Engines	HTML5 Canvas 2D Context API (Ambient Meteo Particles, Radar Sweep, Topographic Mesh, UAV FLIR HUD)
Geospatial Libraries	Leaflet.js v1.9.4, OpenStreetMap Tile Layer, ESRI World Imagery Satellite Layer
External Meteorological APIs	Open-Meteo Global Forecasting, Open-Meteo Air Quality (US AQI), Solar UV Index API
Audio & Voice Engine	Web Audio API (OscillatorNode Dual-Frequency Siren & VHF Beep), Web Speech Synthesis API
Offline & PWA Infrastructure	Service Worker (sw.js), Cache Storage API, HTML5 Web Storage (LocalStorage Buffer)

CATEGORY	SPECIFICATION / TECHNOLOGY STACK
Minimum Client Hardware	Any modern web browser (Google Chrome, Microsoft Edge, Mozilla Firefox, Safari) on Mobile/PC

8. EXPECTED OUTCOMES & SOCIETAL IMPACT

- **Zero-Casualty Target:** Early thermodynamic lightning warnings and proactive village risk mapping minimize preventable deaths during severe monsoons.
- **Uninterrupted Disaster Governance:** DDMA officials and NDRF battalion commanders can operate without cellular dependency using the offline queue and alternative VHF radio grid.
- **Community Empowerment:** Gram Pradhans and rural citizens gain access to single-click WhatsApp advisory broadcasts and emergency concrete shelter locations.

9. CONCLUSION & PROJECT REFERENCES

SkyCast AI demonstrates how artificial intelligence, geospatial intelligence, and client-side resilience engineering can revolutionize district disaster management. By eliminating external API costs, providing seamless bilingual accessibility, and introducing military-grade drone telemetry, the system stands as an exemplary, deployment-ready technological contribution.

References:

- [1] National Disaster Management Authority (NDMA) — *Guidelines on Management of Floods & Urban Inundation*.
- [2] World Meteorological Organization (WMO) — *Convective Available Potential Energy (CAPE) and Severe Weather Forecasting*.
- [3] Open-Meteo Documentation — *High-Resolution Meteorological and Air Quality Data Protocols*.
- [4] Leaflet.js Documentation — *Mobile-Friendly Interactive Geospatial Mapping API*.
- [5] Project Live Deployment: <https://akarsh32-hub.github.io/SkyCast-AI/> | Source: <https://github.com/akarsh32-hub/SkyCast-AI>